

LAMP SHELLS

Lampshells look very like bivalve molluscs, but feed instead with tentacles. They belong to a different, very ancient phylum.

The soft body of a lampshell is enclosed within a two-valved shell that has one valve larger than the other. The shell is either attached to the seabed with a rubbery stalk or is directly cemented to rock. The shell valves are positioned top-and-bottom around the animal – not around the sides, as in bivalve molluscs. Like molluscs, lampshells have a fleshy mantle attached to the inner surface of the shell, which encloses a mantle cavity. In the mantle cavity there is a ring of tentacles with microscopic beating hairs that drive particles to a central mouth – just like the feeding structure in bryozoans.

The fossil record shows that lampshells were much more common and demonstrated much greater diversity in the warm, shallow seas of the Paleozoic Era. They began to decline dramatically during the time of the dinosaurs – perhaps because bivalve molluscs were more successful.

PHYLUM	BRACHIOPODA
CLASSES	3
ORDERS	5
FAMILIES	About 30
SPECIES	414



2–3 cm
¾–1½ in

EUROPEAN LAMP SHELL
Terebratulina retusa
F: Cancellothyrididae

Found from the N.E. Atlantic to the Mediterranean, this lampshell has pear-shaped valves and attaches to vertical rocks using a short stalk.



3–5.5 cm
1¼–2¼ in

hinged, two-valved shell

PACIFIC LAMP SHELL
Terebratalia transversa
F: Terebrataliidae

An abundant lampshell of the northern Pacific, this short-stalked species has a variable smooth or ribbed shell.

1–1.5 cm
¾–½ in



INARTICULATED LAMP SHELL
Novocrania anomala
F: Craniidae

A brachiopod of North Atlantic waters, this species cements its shell to rock, and superficially resembles a limpet.

MOLLUSCS >>

Molluscs are a large, highly diverse group, ranging from blind filter-feeding bivalves cemented to rock, through voraciously grazing snails and slugs, to lively, intelligent octopus and squid.

A typical mollusc has a soft body carried on a large muscular foot, and a head that senses the world using eyes and tentacles. The viscera (internal organs) are contained in a visceral hump, which is covered by a fleshy mantle. This mantle overhangs the edge of the hump, creating a groove called the mantle cavity, which is used for breathing. In most species, this mantle also secretes substances used to make the shell. Most molluscs feed with a tongue called a radula. Coated in teeth made of chitin, the radula moves forwards and backwards through the mouth to rasp at food. Bivalves lack a radula. They feed instead by siphoning water through their shell. Most species trap particles of food in the mucus on their gills.

PHYLUM	MOLLUSCA
CLASSES	9
ORDERS	53
FAMILIES	609
SPECIES	71,719



These limpets have grazed the rock around them bare, but they cannot reach the green algae growing on top of their own shell.

WITH AND WITHOUT SHELLS

A shell is not only a refuge from predators: it is also protection from drying out. Some snails even have a trapdoor called an operculum to seal the opening of the shell. The shell is hardened with minerals obtained from the diet and surrounding water. It is coated in tough protein; and smooth on the inside, to allow the body to slide in and out – some groups have a lining of mother-of-pearl. Many shell-less molluscs defend themselves with chemicals that make them distasteful, or even poisonous, and advertise this with striking colours.

Most cephalopods – tentacled molluscs, such as octopus and squid – lack an obvious shell. Members of this group are entirely predatory and have a horny beak for chewing on flesh. Their muscular foot is modified into their characteristic tentacles, which are used for grasping and swimming.

GETTING OXYGEN

Most molluscs are aquatic and so they have gills for breathing, usually projecting into the mantle cavity and irrigated by a current of water. In most land snails and slugs the mantle cavity is filled with air and functions as a lung. Because many freshwater snails probably evolved from this group, they have a lung too, and must frequently surface to breathe.

A PLACOPHORANS

More worm-like than mollusc-like, the Aplacophora are small cylindrical burrowers that feed on detritus or other invertebrates in the sediment of deep oceans. They lack a shell, but their clearly molluscan features include a rasping, tongue-like radula. The mantle cavity is reduced to an opening at the rear of the body, into which the animal discharges waste from the gut.



3 mm–8 cm
⅛–3¼ in

COMMON GLISTENWORM
Chaetoderma sp.
F: Chaetodermatidae

This worm-like mollusc is covered by a tough cuticle. It burrows through muddy sediment in the depths of the North Atlantic.